INCH-POUND

MIL-DTL-52543E
21 June 2010
SUPERSEDING
MIL-L-52543D
29 March 1989

# DETAIL SPECIFICATION

# LIGHT, MARKER, GROUND-OBSTRUCTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a self-contained, battery-operated, steady or flashing, ground obstruction marker light.

1.2 Classification. The marker lights are of the following battery types, color filter dome sizes, and carrying bag options, as specified (see 6.2).

1.2.1 Batteries. The types of battery are as follows:

Type L - One lantern battery
Type C - Four C-cell batteries

1.2.2 Color filter domes. The color filter dome sizes are as follows:

Size 1 - Small color filter domes in accordance with figure 3
Size 2 - Large color filter domes in accordance with figure 4

1.2.3 Carrying bag options. The options of a carrying bag are as follows:

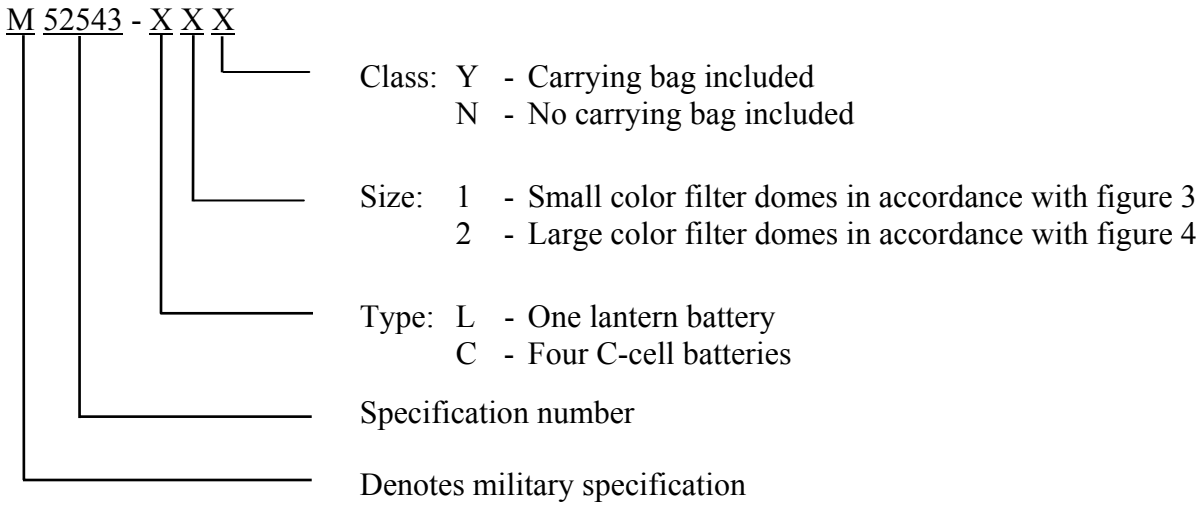
Class Y - Carrying bag included
Class N - No carrying bag included

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1.3 Part or identifying number (PIN). The PIN to be used for a marker light acquired to this specification is created as follows:




## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

| | |
|---|---|
| FED-STD-595/13538 | - Yellow, Gloss |
| FED-STD-595/23538 | - Yellow, Semigloss |
| FED-STD-595/33538 | - Yellow, Flat or Lusterless |

COMMERCIAL ITEM DESCRIPTIONS

| | |
|---|---|
| A-A-52506 | - Clamps, Hose |
| A-A-59708 | - Cloth, Cotton, Aftertreated Water- Repellant, Water-Repellant, and Mildew Resistant, Mildew Resistant |



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DEPARTMENT OF DEFENSE SPECIFICATIONS

| | |
|---|---|
| MIL-P-501 | - Pin, Tent, Metal |
| MIL-W-5625 | - Webbing, Textile, Nylon, Tubular |
| MIL-DTL-7989 | - Covers, Light-Transmitting, for Aeronautical Lights, General Specification for |
| MIL-C-10799 | - Cloth, Coated, Cotton, Vinyl Coated, Fire and Mildew Resistant |
| MS27977 | - Fasteners, Snap, Style 1 (Large Curtain Type) |
| MIL-DTL-32072 | - Thread, Polyester |
| MIL-C-43375 | - Cloth, Duck, Nylon, 12.5 Ounce |

DEPARTMENT OF DEFENSE STANDARDS

| | |
|---|---|
| MIL-STD-130 | - Identification Marking of U.S. Military Property |
| MIL-STD-810 | - Environmental Engineering Considerations and Laboratory Tests |
| MIL-STD-889 | - Dissimilar Metals |

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents. The following other government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

| | |
|---|---|
| Rule 442 | - Usage of Solvents |

(Copies of this document are available online at http://www.arb.ca.gov/ or from Sacramento Metro AQMD, 777 12th Street, Third Floor, Sacramento, CA 95814-1908.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

| | |
|---|---|
| ANSI C18.1M, Part 1 | - American National Standard for Portable Primary Cells and Batteries with Aqueous Electrolyte - General and Specifications |



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ANSI - Continued

| | |
|---|---|
| ANSI C18.1M, Part 2 | - American National Standard for Portable Primary Cells and Batteries with Aqueous Electrolyte - Safety Standard |

(Copies of these documents are available online at http://www.ansi.org/ or from American National Standards Institute, 25 West 43rd Street, 4th Fl., New York, NY 10036.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

| | |
|---|---|
| ASQ Z1.4 | - Sampling Procedures and Tables for Inspection by Attributes |

(Copies of this document are available online at http://www.asq.org/ or from American Society of Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

ASTM INTERNATIONAL

| | |
|---|---|
| ASTM B209 | - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate |
| ASTM D256 | - Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics |
| ASTM D4673 | - Standard Classification System for Acrylonitrile-Butadiene-Styrene (ABS) Plastics and Alloys Molding and Extrusion Materials |
| ASTM D6193 | - Standard Practice for Stitches and Seams |

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

| | |
|---|---|
| SAE AS25050 | - Colors, Aeronautical Lights and Lighting Equipment, General Requirements for |

(Copies of this document are available from http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Description. The marker light (hereinafter referred to as "light") shall consist of a light assembly, a ballast bag and four anchor pins (see 6.5).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor, shall be of the lightest practicable weight suitable for the purpose intended, and shall be subject to all provisions of this specification. The material shall be free from all defects and imperfections that may affect the quality and reliability of the finished product. Lights fabricated of materials other than those specified herein, and any design changes required due to use of such materials, shall be subject to approval of the contracting officer.

3.3.1 Materials deterioration prevention and control. The light shall be fabricated from the compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the light may be exposed. Plastic inherently susceptible to deterioration in the ultraviolet light shall be treated (i.e., ultraviolet stabilized) for the effects of ultraviolet light exposure, except for plastics whose functions would be adversely affected by such treatment. When specified (see 6.2), a certification of ultraviolet treatment to plastic components shall be provided.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contract with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of metals and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the light.

3.4 Construction. The light shall be constructed so that parts will not work loose, and replacement of the battery, dome, deflector, and lamp can be made by personnel without the use of special tools.

3.5 Operation. With the light switch (see 3.6.5) in the "ON" position, the light shall emit a steady glow. With the switch in the "OFF" position, the light shall be completely extinguished. With the switch in the "FLASH" position, the light shall flash in accordance with the flashing frequency and on-time as specified in 3.6.6.



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3.5.1 Operational performance. When tested as specified in 4.4.2.2, the light shall operate as specified in 3.5 without damage (see 6.4.1).

3.5.2 Impact. When tested as specified in 4.4.2.3, the light shall withstand a 6-foot drop without damage.

3.5.3 Rain resistance. When tested as specified 4.4.2.4, the light shall operate in the "FLASH" position as specified in 3.5, without damage and without any noticeable variation in light output. There shall be no noticeable evidence of water penetration inside the canister or dome.

3.5.4 Low-temperature and high-temperature operation. When tested as specified in 4.4.2.5, the light shall operate as specified in 3.5 without damage between 32 °F (0 °C) and 131 °F (55 °C), and the flasher cycle rate shall not change more than ±10 percent between 32 °F (0 °C) and 131 °F (55 °C).

3.5.5 Endurance (operational reliability). When tested as specified in 4.4.2.6, the light shall operate in the "FLASH" position as specified in 3.5 without damage for a period of 50 hours (except for change of lamp) at 90 °F (32.2 °C).

3.6 Light assembly. The light assembly (see 6.5 and figure 2) shall consist of a cylindrical canister, deflector, clear dome, lamp, three-way switch, flasher, and such wiring and fasteners required to make a complete operational assembly when operated with an ANSI C18.1M, Part 1 and Part 2 designated: 6-volt alkaline lantern battery 908A, or 908AC; 6-volt carbon-zinc lantern battery 908CD, or 908D; four 1.5-volt alkaline C-cell batteries 14A, or 14AC; or four 1.5-volt carbon-zinc C-cell batteries 14CD, or 14D. Batteries shall not be furnished with the light. A battery holder of commercial grade plastic to accommodate four C-cells shall be furnished on designs requiring a 6-volt lantern battery.

3.6.1 Canister. The canister, including any canister cap, shall be made of high-impact plastic with a minimum Izod impact strength of 1.5 foot pounds/inch of notch (0.25 inch (0.64 cm) bar) at 73.4 ±3.6 °F (23 ±2 °C), when tested in accordance with method A of ASTM D256. The canister shall contain the battery and electrical circuitry, and shall be compatible with the deflector and dome. A guard on the exterior of the canister shall be provided for the protection of the three-way switch. When color filter domes are furnished in accordance with figure 3 (see 3.6.3), there shall be a space in the canister to store these domes without interference to the operation of the light.

3.6.2 Deflector. The deflector shall be either: (1) plastic with nickel-chrome finish; (2) cold-rolled steel with nickel-chrome finish; or (3) aluminum conforming to ASTM B209, alloy 3003 with a polished anodic oxide finish. The deflector shall contain the light socket for the lamp (see 6.5 and figure 2).

3.6.3 Domes. The clear dome shall conform to figure 4 and shall be made of non-diffusing clear plastic conforming to MIL-DTL-7989, class D. An opaque dome shall be furnished as an accessory and shall conform to figure 4 and be made of plastics conforming to



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MIL-DTL-7989. Red, green, blue, and amber filter domes shall be furnished as accessories and shall conform to either figure 3 or figure 4. The chromaticity and transmittance of the color filter domes shall be as specified in SAE AS25050, type I. When color filter domes are furnished in accordance with figure 3, they shall be stored within the canister (see 3.6.1), and when stored, shall not interfere with the operation of the light. The color filter domes conforming to figure 3 shall be made to fit over the lamp and maintain their position over the lamp while the light is inverted and shook (see 4.4.2.7). Also, when tested as specified in 4.4.2.5, the blue filter dome conforming to figure 3 shall not soften or deform when installed over the lamp and when the light is operated in the steady "ON" mode for 6 hours at an ambient air temperature of 131 °F (55 °C).

3.6.4 Lamp. The lamp shall be a commercial-type lamp, manufacturer's trade number 503. Two spare lamps shall be provided with each light (see 6.5 and figure 2).

3.6.5 Switch. An environmentally-sealed, externally-operated toggle switch shall be furnished. The switch shall provide "ON", "OFF", and "FLASH" positions. The positions shall be labeled "ON", "OFF", and "FLASH" respectively. The label characters shall be 0.18 inches tall and of a color contrasting with the background. A neoprene or silicone rubber boot with a nickel-plated brass mounting-nut shall be provided to fit over the switch and prevent moisture from entering the switch and canister.

3.6.6 Flasher. The flasher or flasher circuit shall provide a flashing frequency of 60 cycles ±10 cycles per minute, and the on-time for the lamp shall be not less than 0.3-second duration for each flash. Cycle rate shall not change more than ±10 percent at ambient temperatures between 32 °F (0 °C) and 131 °F (55 °C).

3.6.7 Waterproofing gaskets. Gaskets used to seal the light from moisture shall be affixed to one of the components (see 6.5 and figure 2).

3.7 Ballast bag. The ballast bag (see 6.5 and figure 5) shall be made of either nylon duck cloth conforming to MIL-C-43375, class 2 or cotton duck cloth conforming to MIL-C-10799, type II, class 3. The cotton duck cloth shall be treated with one of the following mildew inhibitors in accordance with A-A-59708:

a. 2,2'-Methylenebis(4-chlorophenol) with a 1.10 to 1.90 percent total weight of finished cloth, 2 baths with aqueous or solvent application.

b. Copper 8-quinolinolate in dispersion form with a 0.18 to 0.27 percent total weight of finished cloth, 1 bath with aqueous or solvent application.

3.7.1 Thread. Thread conforming to MIL-DTL-32072, type I, class I, shall be used for all stitches. Stitches shall conform to ASTM D6193, type 301 with 5 to 7 stitches per inch (2.54 cm).

3.7.2 Opening. The ballast bag shall be provided with one 12 ±0.5 inch (30.48 ±1.27 cm) opening flap, which shall extend up from the base as shown in figure 1, for filling and



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emptying the bag. The opening flap shall be provided with three snap fasteners conforming to MS27977-28B for the socket, long prong; MS27977-3B for the clinch plate; MS27977-8B for the stud, eyelet base; and MS27977-10B for the washer. All fasteners shall be corrosion resistant with a flat black finish. The ballast bag shall be marked "FILL WITH BALLAST BEFORE USING", in black letters not less than 0.375 inch (0.95 cm) high.

3.7.3 Holddown lines. Two nylon lines 0.5 inch (1.27 cm) in diameter conforming to MIL-W-5625 shall be stitched diagonally from corner to corner to the bottom of the ballast bag. The nylon lines shall not overlap where the lines intersect at the bottom center of the ballast bag (see figure 5). Each end of the holddown lines shall be provided with a 2-inch (5.08-cm) diameter stitched loop. The four loops shall project 6 inches (15.24 cm) ±0.5 inch (1.27 cm) from the bottom corners of the ballast bag.

3.7.4 Collar. On designs requiring a collar, the collar shall be provided with a clamp conforming to A-A-52506, type F, corrosion-resisting steel, 4.5-inch (11.43 cm) nominal clamp size with thumb screw adjusting mechanism. The clamp shall pass through a minimum of two stitched loops (see 6.5 and figure 1).

3.8 Anchor pins. Four anchor pins, each 9 inches (22.86 cm) in length, shall be provided with each light. Pins shall conform to MIL-P-501, type I, aluminum (except that the color shall be as specified in 3.14) or ASTM D4673, class 2, high impact plastic, similar in construction to the type I aluminum pin.

3.9 Carrying bag. When specified (see 6.2), a nylon or cotton duck cloth carrying-bag shall be provided for carrying a set of eight lights. The nylon duck cloth shall conform to MIL-C-43375, class 2. The cotton duck cloth shall conform to MIL-C-10799, type II, class 3. The cotton duck cloth shall be treated with one of the following mildew inhibitors in accordance with A-A-59708:

a. 2,2'-Methylenebis(4-chlorophenol) with a 1.10 to 1.90 percent total weight of finished cloth, 2 baths with aqueous or solvent application.

b. Copper 8-quinolinolate in dispersion form with a 0.18 to 0.27 percent total weight of finished cloth, 1 bath with aqueous or solvent application.

3.9.1 Thread. Thread conforming to MIL-DTL-32072, type I, class I, shall be used for all stitches. Stitches shall conform to ASTM D6193, type 301 with 5 to 7 stitches per inch (2.54 cm). The carrying bag shall be provided with two carrying handles or straps.

3.9.2 Opening. An opening shall be provided for installing the set of eight marker lights. The opening shall be provided with snap fasteners conforming to MS27977-28B for the socket, long prong; MS27977-3B for the clinch plate; MS27977-8B for the stud, eyelet base; and MS27977-10B for the washer. All fasteners shall have a flat black finish. The carrying bag shall be marked "CARRYING BAG FOR MARKER LIGHTS" in black letters with a 0.5 inch (1.27 cm) minimum height.



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3.10 Interchangeability. All parts having the same contractor's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.11 Identification marking. The light shall be indentified in accordance with MIL-STD-130.

3.12 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with Simpson Coatings fungus-resistance varnish 101-004, or equivalent:

a. Components or circuit elements that are inherently fungus-and-moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.1 Composition. All applications of varnish shall fully comply with SCAQMD Rule 442.

3.13 Dimensions. The dimensions of the light, with 20 pounds (9 kg) of sand as ballast, shall not exceed the following:

Length - 12.5 inches (31.75 cm)
Width - 12.5 inches (31.75 cm)
Height - 11.5 inches (29.21 cm)

3.14 Color. The color of the canister, ballast bag, thread, holddown lines, anchor pins, and carrying bag shall be aviation yellow numbers 13538, 23538, or 33538 as specified in FED-STD-595.

3.15 Workmanship. The light, including all parts and accessories shall be fabricated and finished to ensure freedom from blemishes, defects, burns, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts, and tightness of assembled components.



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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. The light shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests. The light shall be tested as specified in 4.4.2. Failure of any test shall be cause for rejection.

4.3 Conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be determined by using ASQ Z1.4, with inspection level S-2, with the Acceptance Quality Limit (AQL) as specified in the contract (see 6.2).

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor characteristics for possible defects as specified in 4.4.1.

4.3.3 Tests.

4.3.3.1 Individual. Each light shall be tested as specified in 4.4.2.1 and 4.4.2.2.

4.3.3.2 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.2 through 4.4.2.7. Tests shall be conducted in the sequence shown. Failure of any test shall be cause for rejection of the lot.

4.4 Inspection procedure.

4.4.1 Examination. The lights shall be examined for the following defects:

Major.

101. Parts or components missing or out of place, damaged or not functional (see 3.2).
102. Assembly incomplete or not correct (see 3.2).
103. Materials not as specified (see 3.3).



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104. Materials not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments, and when required, certification of ultraviolet treatment to plastic components not provided (see 3.3.1).
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
106. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the light (see 3.3.2).
108. Construction not as specified (see 3.4).
109. Light assembly not as specified (see 3.6).
110. Canister not as specified (see 3.6.1).
111. Deflector not as specified (see 3.6.2).
112. Domes not as specified (see 3.6.3).
113. Lamp not as specified (see 3.6.4).
114. Switch not as specified (see 3.6.5).
115. Gaskets not as specified (see 3.6.7).
116. Ballast bag not as specified (see 3.7).
117. Holddown lines not as specified (see 3.7.3).
118. When required, collar not as specified (see 3.7.4).
119. Anchor pins missing or not as specified (see 3.8).
120. When required, carrying bag not as specified (see 3.9).
121. Parts not interchangeable (see 3.10).
122. Fungus and moisture resistance not as specified (see 3.12).
123. Anti-fungal varnish composition not as specified (see 3.12.1).
124. Dimensions not as specified (see 3.13).
125. Color not as specified (see 3.14).
126. Workmanship not as specified (see 3.15).

Minor.

201. Identification marking missing or not as specified (see 3.11).

4.4.2 Tests.

4.4.2.1 Test conditions. Unless otherwise specified, test shall be conducted at standard ambient temperature as defined in MIL-STD-810. Prior to each test, a new 6-volt battery or four new 1.5-volt C-cell batteries, as applicable, shall be installed in the light and the light shall be operated in each mode ("ON", "OFF", and "FLASH") to determine that the light meets the requirements of 3.5.

4.4.2.2 Operating. Operate the light in each mode ("ON", "OFF", and "FLASH") five times. Failure to operate in each mode five times without damage shall constitute failure of this test.



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4.4.2.3 Impact. Fill the ballast bag with 20 pounds (9 kg) of sand. Drop the light from a height of 6 feet (1.83 m) upon a hard, rigid surface. Then operate the light in each mode. Evidence of spare lamps that have fallen out of the friction sockets, any damage or inability of the light to operate in each mode as specified in 3.5 after the drop shall constitute failure of this test.

4.4.2.4 Rain. Subject the light to a rain test in accordance with MIL-STD-810, method 506, procedure III, for a minimum of 15 minutes. During the test, the light shall be operating in the "FLASH" mode. Nonconformance to 3.5.3 shall constitute failure of this test.

4.4.2.5 Low-temperature and high-temperature operation. Cold soak the light, without the battery, for 4 hours at 32 °F (0 °C). Install the battery and operate the light for 15 minutes in the "FLASH" mode. At the end of 15 minutes, return the light to ambient temperature and repeat the test specified in 4.4.2.2. Then hot soak the light, without the battery, for 4 hours at the 131 °F (55 °C). Install the battery and operate the light for 15 minutes in the "FLASH" mode. When a blue filter in accordance with figure 3 is furnished (see 3.6.3), install this filter in place of the clear dome, install a new battery, and operate the light in the steady "ON" mode for six hours at 131 °F (55 °C). Return the light (with either the clear dome or blue filter, as applicable) to ambient temperature and repeat the test specified in 4.4.2.2. Nonconformance to 3.5.4 and 3.6.3, as applicable, shall constitute failure of this test.

4.4.2.6 Endurance. Install a new 6-volt battery or four new 1.5-volt C-cell batteries, as applicable, in the light. Operate the light continuously in the "FLASH" mode for not less than 50 hours in an ambient temperature of 90 °F (32.2 °C). Nonconformance to 3.5.5 shall constitute failure of this test.

4.4.2.7 Color filter dome adhesion. When applicable (see 3.6.3), invert the light assembly and shake vigorously. Nonconformance to 3.6.3 shall constitute failure of this test.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The light covered by this specification is intended for use as an emergency light for applications requiring portability and short-range visibility from the air, such as marking closed runways, helicopter operations, soft spots, or runway lighting on aircraft landing strips.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Battery type, color filter dome size, and carrying bag option required (see 1.2).
c. When first article is required (see 3.1).
d. When a certification of ultraviolet treatment to plastic components is required (see 3.3.1).
e. AQL (see 4.3.1).
f. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition. The following definition should apply throughout this specification.

6.4.1 Damage. Damage is deformation, loosening, separation of the canister from the ballast bag, breakage, change of fit of any component or part, corrosion, or any condition resulting in malfunction of the marker light or any component or part.

6.5 Information figures. Figures 1, 2, and 5 show a type of marker light, which has been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other marker lights that conform to this specification.

6.6 Dimensional figures. Figures 3 and 4 show dimensions for the applicable dome that must be achieved (see 3.6.3).



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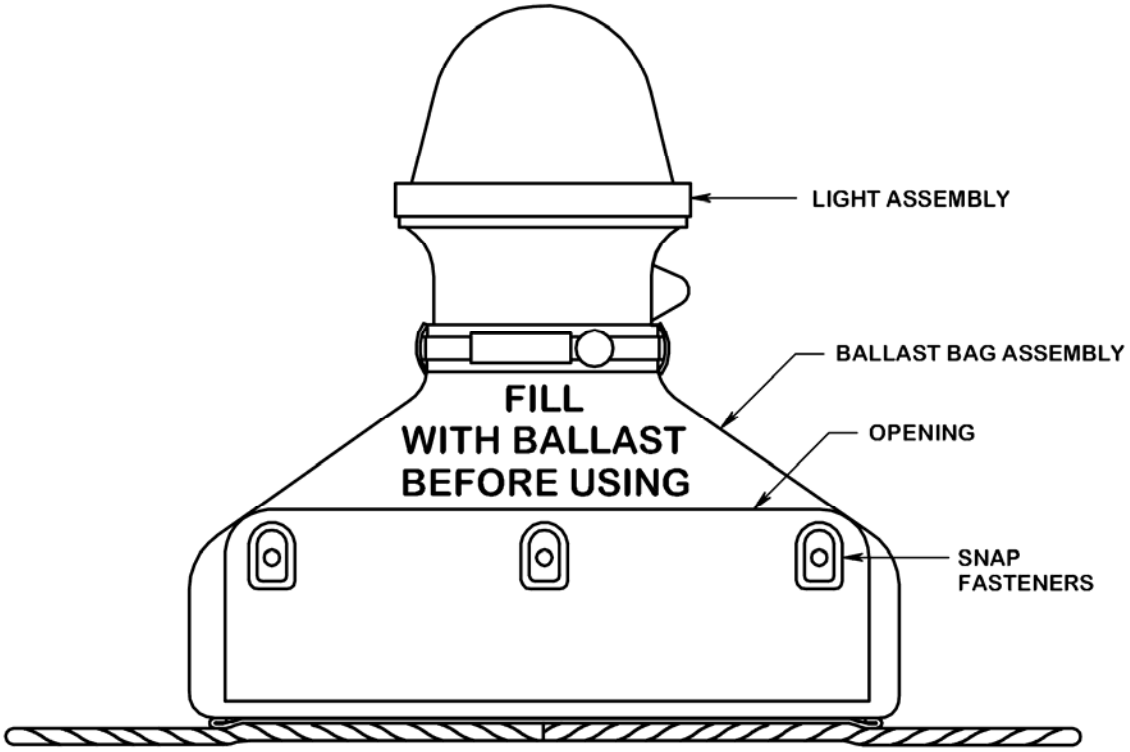
6.7 Subject term (key word) listing.

Ballast bag
Blinking light
Emergency light
Flasher
Flashing light
Light assembly
Portable light
Strobe light

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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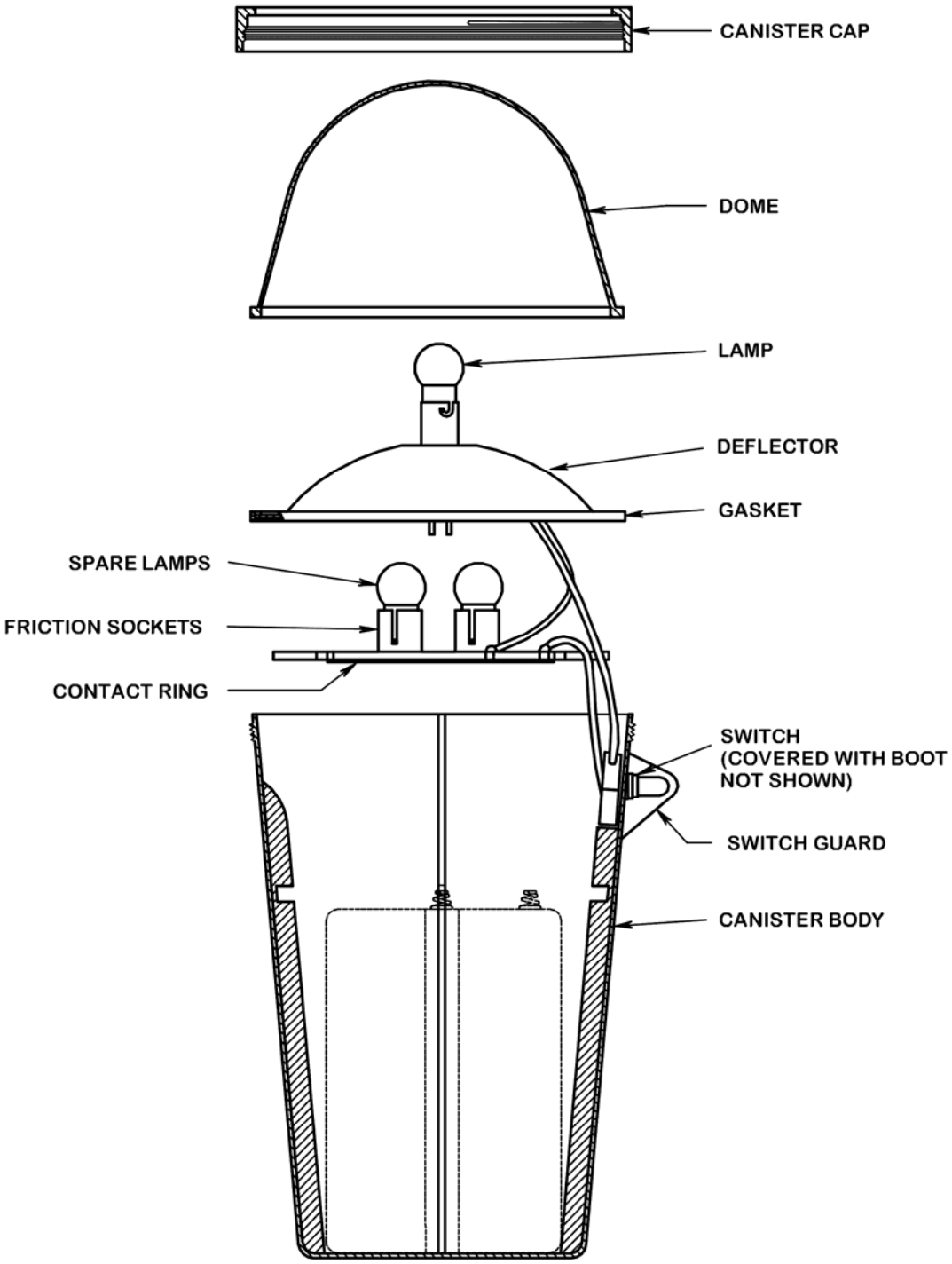



NOTE:
1. Anchor pins not shown.

FIGURE 1. Light, marker, ground obstruction.



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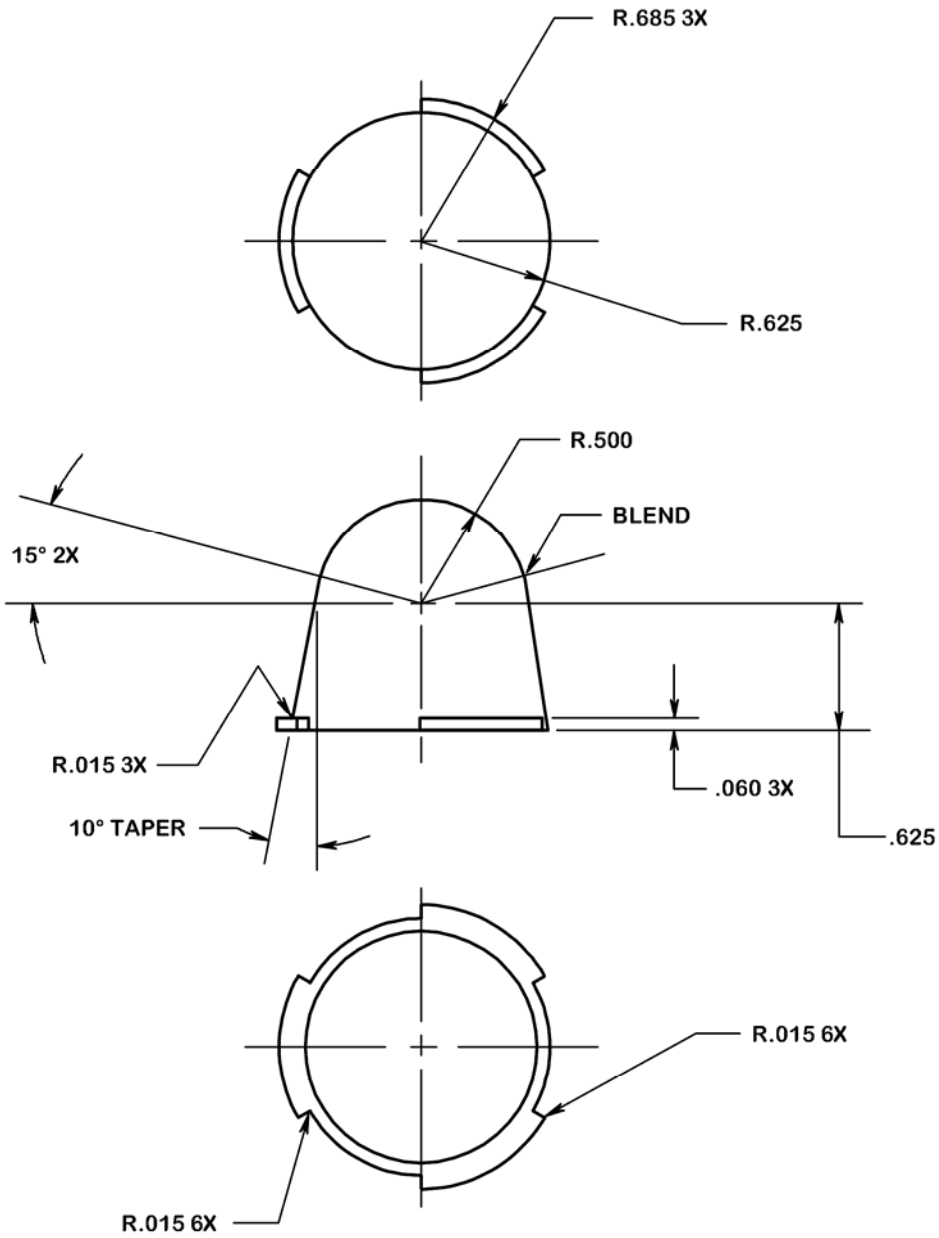



NOTE:
1. Light shown can use either a 6-volt lantern battery or four C-cell batteries in a battery holder. Batteries are not furnished with the light.

FIGURE 2. Light assembly.



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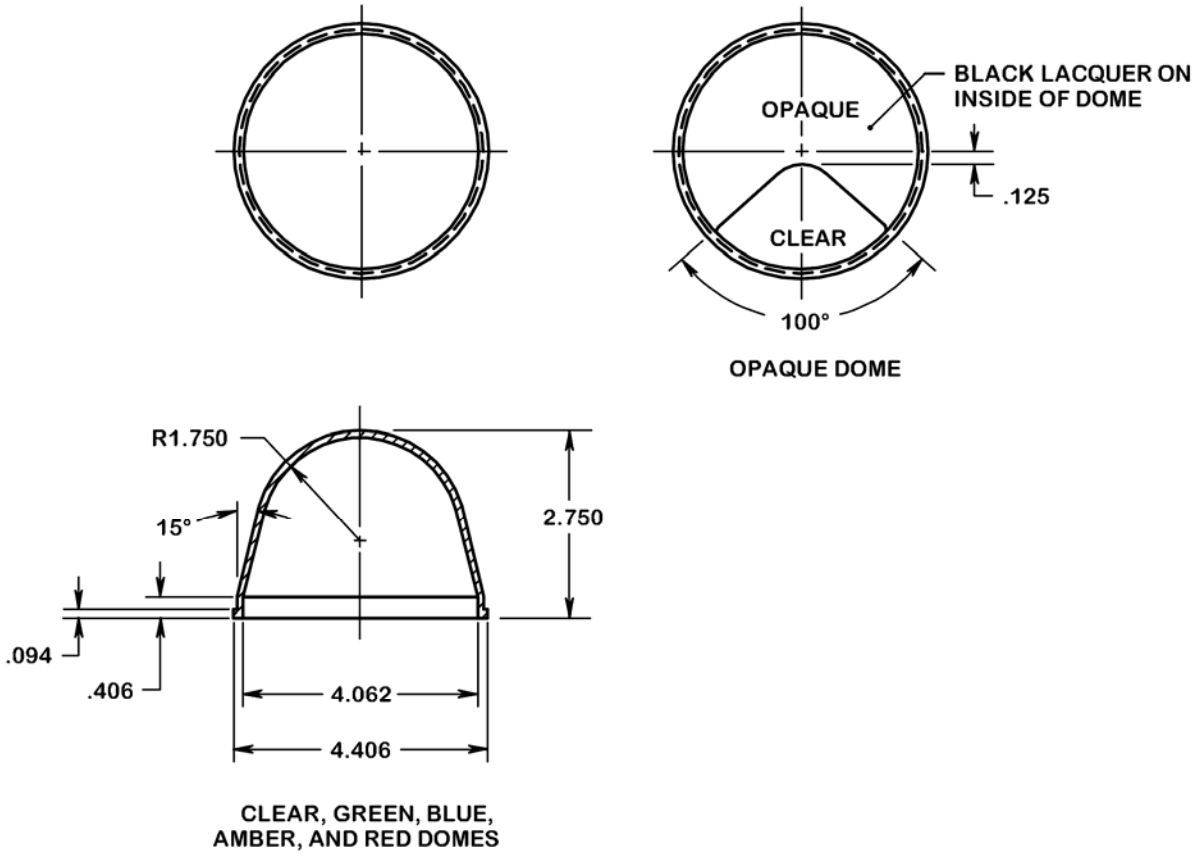



NOTES:
1. All dimensions are in inches.
2. Nominal wall thickness is .060.
3. Unless otherwise specified, tolerances are ±.005, and angles are nominal.
4. Green, blue, amber, and red filter domes.

FIGURE 3. Filter domes, molded, small, color.



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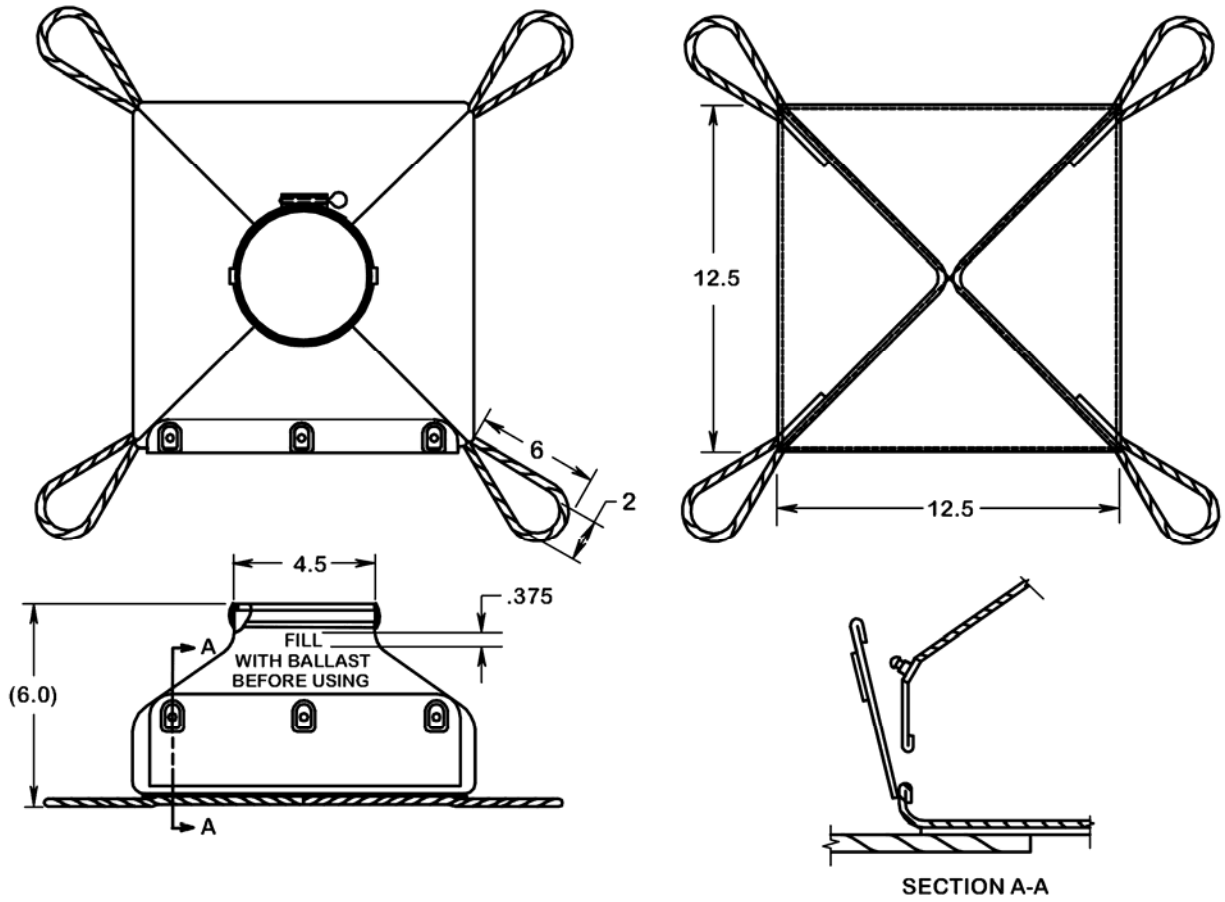



NOTES:
1. All dimensions are in inches.
2. Unless otherwise specified, tolerances are: angles ±1°, decimals ±.031.
3. Color filter domes, opaque dome, and clear dome have same dimensions.

FIGURE 4. Filter domes, molded, large, clear, opaque and color.



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NOTE:
1. All dimensions are in inches.

FIGURE 5. Bag, ballast assembly.



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Custodians:
Army - CR4
Navy - AS
DLA - GS

Preparing Activity:
DLA - GS2

(Project 6230-2010-003)

Review Activities:
Army - AV
Navy - YD
DLA - IS

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